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Real-World Robot Control Based on Contrastive Deep Active Inference with Demonstrations

KENTARO FUJII[1], TAKUYA ISOMURA[2], and SHINGO MURATA[1]
[1]Graduate School of Integrated Design Engineering, Keio University, Kanagawa, Japan (e-mail:oakwood.n14.4sp@keio.jp, murata@elec.keio.ac.jp)
[2]Brain Intelligence Theory Unit, RIKEN Center for Brain Science, Saitama, Japan (e-mail: takuya.isomura@riken.jp)

Corresponding author: Shingo Murata (e-mail: murata@elec.keio.ac.jp).

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ABSTRACT Despite significant advances in robotics and deep learning, the ability of robots to perceive and act remain far below that of humans. To bridge this gap, we utilize active inference, a framework based on the free-energy principle that accounts for various human brain functions. Despite the utility of active inference, most previous studies have focused on simulated or real-world environments with low-dimensional observations. Therefore, direct applications in real-world robotics with high-dimensional observation and action spaces remain a challenge. To address this issue, we combine the active inference framework with contrastive learning and develop an extension by introducing learning from demonstration. This framework comprises a world model, an action model, and an expected free-energy model. The world model ensures an adequate perception of the environment by minimizing contrastive variational free energy. Meanwhile, the expected free-energy model estimates the contrastive expected free energy, while the action model generates adaptive actions by learning from demonstrations, with the aim of minimizing the estimates provided by the expected free-energy model. We show the effectiveness and robustness of the proposed framework through image-based robotic reaching tasks in both learned and unlearned real-world environments. As a result, the proposed framework outperformed recent learning-from-demonstration baselines. Moreover, the proposed framework realized performance improvements by leveraging estimated expected free energy for action selection. These results highlight the utility of active inference with contrastive learning and learning from demonstration for real-world robot control.

INDEX TERMS Active inference, robot learning, learning from demonstration, world model, free energy principle, contrastive learning, imitation learning

I. INTRODUCTION

ROBOTS capable of human-like perception and action are highly desired for various real-world applications but remain challenging to develop. To overcome this challenge, deep learning approaches to robot control have been developed, including "learning from demonstration (LfD)," which aims to imitate the behavior of experts [1]–[3], and deep "reinforcement learning (RL)," which aims to maximize future rewards [4], [5]. In these approaches, robots utilize data collected from expert demonstrations [1], [3], [6] or throughout the robot's own exploration [4], [5], [7]. To enhance the robot's perception and action capabilities, it may be possible to utilize an approach based on the computational principles of the human brain [8]–[10]. In this work, we focus on active inference, a framework based on the free-energy principle that accounts for various brain functions [11], [12]. For clarity, the abbreviated terms and their descriptions, which will appear later in the paper, are summarized in Table I in advance.

In active inference, both perception and action are framed with the objective of minimizing free energy [11], [12], [17]. Agents are assumed to model the external world in their internal states, often referred to as the "world model" [18], [19]. Agents update their world model to minimize variational free energy and enhance appropriate perception. Furthermore, agents select actions that satisfy their preferred outcome, by minimizing free energy in the future, which is called expected free energy (EFE). Recently, deep active inference has provided a framework to scale up active inference with deep learning [20]–[22]. In deep active inference, agents are widely learned to minimize the free energy calculated through likelihood-based approaches involving reconstruction of the high-dimensional observation (e.g., images) [18], [21], [23]. However, in likelihood-based approaches, an agent's infer-

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TABLE 1. Glossary of Terms

Term	Description
LfD	Learning from demonstration.
RL	Reinforcement learning.
EFE	Expected free energy.
KLD	Kullback-Leibler divergence.
NCE	Noise contrastive estimation, as detailed in (6).
RSSM	Recurrent state-space model, as detailed in Section IV-B1.
GRU	Gated recurrent unit.
CNN	Convolutional neural network.
KL-balancing	The technique used for training RSSM, as detailed in (11). For more details, please refer the original paper [13].
MLP	Multilayer perceptron.
AM	Action model.
EFEM	EFE model.
AM action	One of the methods to generate action using the action model, as detailed in Section IV-C.
EFEM action	Another method to generate action using the action and EFE model, as detailed in Section IV-C and Algorithm 1, 2.
CNMP	Conditional neural movement primitives. One of the baselines in this paper, explained in Section V-B. For details, please refer the original paper [14].
CNEP	Conditional neural expert processes. One of the baselines in this paper, explained in Section V-B. For details, please refer the original paper [15].
IBC	Implicit behavior cloning. One of the baselines in this paper, explained in Section V-B. For details, please refer the original paper [16].

ence is strongly influenced by the size of observations on the pixel level, which tends to overlook small yet significant objects or behaviors [24], [25].

Meanwhile, contrastive active inference [25] introduces contrastive learning into free energy, proposing contrastive free energy and provides an approach that addresses the challenges of likelihood-based approaches [26]. Contrastive learning is a feature-extraction method that trains the model so that similar data pairs have closely aligned features, whereas dissimilar pairs have distinctly separate features [27]–[29]. Although contrastive active inference shows promise, the relationship between contrastive and original free energies remains to be fully explored. Furthermore, contrastive active inference involves exploration-driven data collection, which carries the risk of damaging the robot or environment, making direct application to real-world robots challenging [30].

To address these challenges, we propose a novel framework for applying contrastive active inference to real-world robot control by incorporating LfD. We delve into the relationship between contrastive and original free energies and then construct a framework consisting of a world model, an action model, and an EFE model. We train this framework using demonstration data collected from demonstrations by human experts. Through experiments, we demonstrate the effectiveness of the proposed framework for real-world robot control, showing that robots can achieve adaptive behavior by selecting the action with the lowest EFE.

The remainder of this paper is structured as follows. In section II, we review related work, and then explain active inference and free energy in section III. We present the proposed framework in section IV. Section V provides an explanation of the experimental environment. In section VI, we present and discuss the experimental results. Finally, we draw conclusions and consider future directions in section VII.

II. RELATED WORK

A. ROBOT LEARNING

Broadly, there are two approaches to using deep learning for robot: RL [1], [7], [8] and LfD [1], [2], [31]. RL presents a challenge for real-world applications because RL frameworks often employ exploration-driven data collection [7], [30]. During exploration, especially in the early stages, the robot often performs movements unrelated to accomplishing its task. As a result, the robot has an increased risk of colliding with objects in its environment [7]. Efforts to circumvent this challenge include implementing effective exploration strategies [5], using offline RL [4], [32], and integration with LfD [30].

LfD enables the robot to learn by imitating demonstrations performed by human experts [1], [2], [30]. This learning process is driven by matching the distribution of actions taken by the robot with that of the expert [1]. Demonstrations are collected under human supervision and involve closely monitoring the robot's training process. As the robot learns to generate actions that mimic these demonstrations, it becomes easier to keep the actions within the intended environmental range. This significantly enhances the safety of both the robots and their operational environment [1], [30].

To enhance robot learning with LfD, previous research focused on collecting demonstrations as a type of curiosity-driven play by human experts [6], [33], [34] as well as augmenting demonstration data [35], [36] and utilizing human language instructions for planning [31], [37], [38]. In the present work, we utilize LfD to enhance and scale the active inference framework for application in real-world robot control. The proposed framework not only generates actions by imitating demonstrations but also adapts to dynamically changing environments by minimizing EFE.

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B. DEEP LEARNING AND ACTIVE INFERENCE

In active inference, agents need the capability to predict future states for EFE calculation and policy evaluation [12], [17]. The concept of the world model, widely employed in RL [13], [39], [40] and LfD [41], [42], provides the capability to predict future states and observations without actual observations, based solely on policies. While active inference using the world model is promising, most of these studies are framed within simulation environments that handle discrete action space [23], [43], [44], making their direct application to real-world robots challenging [45]. Furthermore, active inference frameworks with continuous action spaces, as applied to real-world robots, often reduce originally high-dimensional observations to lower dimensions [10], [46], [47] with the aim of reducing the computational costs associated with likelihood-based EFE, which involves the reconstruction of observations. Handling high-dimensional observations is essential for robots to adapt to real-world environments.

Contrastive active inference [25] is a framework designed for continuous action spaces and high-dimensional observation spaces. It introduces contrastive learning [48], [49] and extends the original free energy by defining the contrastive free energy. Contrastive learning enables the extraction of features that effectively represent observations without the need to reconstruct observations during learning, removing the requirement to develop a decoder for the observations [27], [29], [50]. By employing contrastive learning, EFE can be calculated without the expensive calculations needed to reconstruct observations [25]. Moreover, although likelihood-based approaches struggle to extract the features of small objects in observations, contrastive learning is capable of extracting features from even small objects, providing effective representations that are beneficial for an agent's learning [24], [25], [51]. Contrastive active inference is a promising approach; however, directly applying it to real-world robots is not straightforward because exploration-driven data collection is unreliable and inefficient [1], [30]. Therefore, we aim to extend contrastive active inference with LfD for real-world robot applications. To this end, we revisit the derivations of contrastive and original free energies in order to more closely couple the derivation of contrastive free energy with that of the original free energy.

III. THE FORMULATION OF ACTIVE INFERENCE

The free-energy principle [11], [12] is a computational principle that accounts for various brain functions. Observations o are generated by hidden states z that cannot be directly observed. These hidden states transform to the next states through actions a given to the world, which evolve over time. This process can be treated as a partially observable Markov decision process [12], [43]. The free energy principle assumes that a common goal of human perception and action is to minimize "surprise" $-\log p(o)$. Although direct minimization of surprise is difficult, minimization of variational free energy, an upper bound on surprise, provides a tractable proxy [12], [52]. The variational free energy $\mathcal{F}(t)$ at a given time t is defined as follows[1] [25]:

$$\mathcal{F}(t) = E_{q(z_t)}\left[\log q\left(z_t\right) - \log p\left(o_t, z_t\right)\right] \geq -\log p\left(o_t\right). \tag{1}$$

Here, $q(z_t)$ denotes the approximate posterior. In this paper, we specify the approximate posterior as the conditional distribution given observation, expressed as $q(z_t|o_t)$. Because the true posterior $p(z_t|o_t)$ is unknown, the hidden states z_t are inferred from observation o_t through Bayesian belief updating under this approximate posterior. The variational free energy $\mathcal{F}(t)$ can also be expressed as follows:

$$\begin{aligned}\mathcal{F}(t) &= D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t\right)\right] - E_{q(z_t|o_t)}\left[\log p\left(o_t|z_t\right)\right] \\ &= D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t|o_t\right)\right] - \log p\left(o_t\right).\end{aligned} \tag{2}$$

In the first line of (2), the variational free energy $\mathcal{F}(t)$ is defined as the negative evidence lower bound, which is known from the variational autoencoder [54], [55]. In the second line of (2), the variational free energy $\mathcal{F}(t)$ is decomposed into a the Kullback–Leibler divergence (KLD) between the approximate posterior and the true posterior as well as a surprise. Given that KLD is non-negative, this shows that the variational free energy $\mathcal{F}(t)$ acts as an upper bound on surprise.

When considering an action, the optimal action is one that minimizes EFE. This introduces the need to consider the agent's future action a [25], [56], expanding the biased generative model $\tilde{p}(o_\tau, z_\tau, a_\tau) = p(a_\tau)p(z_\tau|o_\tau)\tilde{p}(o_\tau)$ and the approximate posterior $q(z_\tau, a_\tau)$, where τ refers to a time step in the future, $\tau \in \{t+1, ..., T\}$ with t reflecting the current time step. Here, $\tilde{p}(o_\tau)$ is called "prior preference," which distributes according to the preferred observation $\tilde{o}$ of the agent [12], [17], [25]. EFE is expressed as follows [25]:

$$\begin{aligned}\mathcal{G}(\tau) &= E_{q(o_\tau, z_\tau, a_\tau)}\left[\log q\left(z_\tau, a_\tau\right) - \log \tilde{p}\left(o_\tau, z_\tau, a_\tau\right)\right] \\ &= E_{p(o_\tau|z_\tau)}\left[D_{\mathrm{KL}}\left[q\left(z_\tau, a_\tau\right)\|p\left(z_\tau, a_\tau|o_\tau\right)\right]\right] \\ &\quad - E_{q(o_\tau)}\left[\log \tilde{p}\left(o_\tau\right)\right] \\ &\approx -E_{q(o_\tau)}\left[D_{\mathrm{KL}}\left[q\left(z_\tau|o_\tau\right)\|q\left(z_\tau\right)\right]\right] \\ &\quad - E_{q(o_\tau)}\left[\log \tilde{p}\left(o_\tau\right)\right] \\ &\quad + E_{q(z_\tau, a_\tau)}\left[\log q\left(a_\tau|z_\tau\right) - \log p\left(a_\tau\right)\right].\end{aligned} \tag{3}$$

Here, the expectation is over predicted observations, states, and actions $q(o_\tau, z_\tau, a_\tau) = p(o_\tau|z_\tau)q(z_\tau, a_\tau)$. We assume that the distribution of actions is not conditioned on observations o_τ. In the third line of (3), we assume $q\left(z_\tau \mid o_\tau\right) \approx p\left(z_\tau \mid o_\tau\right)$. Because observations are not yet acquired, in the future, it will be necessary to set expectations regarding the observations as well. EFE can ultimately be decomposed into three terms. The first term encourages an understanding of the presence of uncertain states and actively obtains observations in order to reduce uncertainty. The second term encourages

[1]The original free energy is dependent on past states [53], but for simplicity, we present a formulation here that accounts for only a single time step [25]. However, in practice, it is implemented to be dependent on past states, as shown in (11).

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actions that better fulfill the agent's preferences. The third term can be seen as the KLD between the action's predictive distribution and the prior. Alternatively, the third term can be interpreted as the entropy $-E_{q(z_\tau)}[\mathcal{H}(q(a_\tau|z_\tau)]$ [25], [57].

For further explanation on active inference, please refer to comprehensive tutorial [52] and survey [58] papers.

IV. METHODS

A. SURPRISE UPPER BOUND

1) Contrastive variational free energy

In active inference, the objective is to minimize an upper bound on surprise $-\log p(o_t)$. In this study, we consider an extended upper bound $F_\alpha(t)$ on surprise $-\log p(o_t)$, which is controlled by the hyperparameter α, as follows:

$$\begin{aligned} F_\alpha(t) &= \alpha \cdot D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t|o_t\right)\right] - \log p\left(o_t\right) \\ &\geq -\log p\left(o_t\right). \end{aligned} \tag{4}$$

When α is set to 1, it corresponds to the original variational free energy $\mathcal{F}(t)$ (2). For sufficiently large values of $\alpha \gg 1$, $-\log p(o_t)$ becomes negligible, and only the KLD between the approximate posterior and the true posterior remains effective, resulting in the contrastive variational free energy $\mathcal{F}_{\mathrm{CL}}(t)$ [25]:

$$\begin{aligned} F_{\alpha\gg1}(t) &\cong D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t|o_t\right)\right] \\ &\leq D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t\right)\right] - I_{\mathrm{NCE}}\left(Z_t;O_t\right) \\ &= \mathcal{F}_{\mathrm{CL}}(t). \end{aligned} \tag{5}$$

Here, I_{NCE}, which is known as noise contrastive estimation (NCE) [59], corresponds to a lower bound on mutual information [49]. Given two random variables X and Y, the NCE lower bound is:

$$I(X;Y) \geq I_{\mathrm{NCE}}(X;Y) \triangleq \mathbb{E}\left[\frac{1}{K}\sum_{i=1}^{K}\log\frac{e^{f(x_i,y_i)}}{\frac{1}{K}\sum_{j=1}^{K}e^{f(x_i,y_j)}}\right]. \tag{6}$$

In this context, during training, expectations are calculated over K independent samples, specifically, mini-batches of X and Y. Here, $f(x,y)$ serves as a critic that approximates $\log\frac{p(x|y)}{p(x)}$ [48], [49]. It functions effectively by taking the inner product of the transformed samples from X and Y, using arbitrary functions h and g such that $f(x,y) = h(x)^T g(y)$ [25], [48]. This derivation shows that the contrastive free energy is an upper bound on the surprise and the original free energy. Accordingly, this derivation is more consistent with the free-energy principle, minimizing the upper bound on surprise$-\log p(o_t)$.

2) Contrastive expected free energy

When considering the control, the upper bound on surprise $F_\alpha(t)$ must be extended to include actions. Accordingly, we define $G_\alpha(\tau)$ as follows:

$$\begin{aligned} G_\alpha(\tau) =& \alpha \cdot E_{p(o_\tau|z_\tau)}\left[D_{\mathrm{KL}}\left[q\left(z_\tau,a_\tau\right)\|p\left(z_\tau,a_\tau|o_\tau\right)\right]\right] \\ &- E_{q(o_\tau)}\left[\log\tilde{p}\left(o_\tau\right)\right]. \end{aligned} \tag{7}$$

When α is set to 1, it corresponds to the original EFE $\mathcal{G}(\tau)$, as in (3). For sufficiently large values of $\alpha \gg 1$, the negative log prior preference $-\log\tilde{p}(o_\tau)$ becomes negligible. In the contrastive active inference framework, there is no decoder, meaning there is no likelihood function for observation $p(o_\tau|z_\tau)$. Instead, $p(o_\tau|z_\tau)$ is replaced by the agent's prior preference $\tilde{p}(o_\tau)$ [25]. We further assume $q(z_\tau|o_\tau) \approx p(z_\tau|o_\tau)$; therefore, $E_{q(z_\tau)}\left[\log q\left(z_\tau\right) - \log p\left(z_\tau\right)\right] = 0$, as in previous studies [25], [60]. The contrastive EFE $\mathcal{G}_{\mathrm{CL}}(\tau)$ [25] is then derived as follows:

$$\begin{aligned} G_{\alpha\gg1}(\tau) &\cong E_{\tilde{p}(o_\tau)}\left[D_{\mathrm{KL}}\left[q\left(z_\tau,a_\tau\right)\|p\left(z_\tau,a_\tau|o_\tau\right)\right]\right] \\ &\approx -E_{\tilde{p}(o_\tau)q(z_\tau)}\left[\log p\left(z_\tau|o_\tau\right) - \log p\left(z_\tau\right)\right] \\ &\quad + E_{q(a_\tau,z_\tau)}\left[\log q\left(a_\tau|z_\tau\right) - \log p\left(a_\tau\right)\right] \\ &\leq -I_{\mathrm{NCE}}\left(Z_\tau;\tilde{O}_\tau\right) + E_{q(z_\tau)}\left[D_{\mathrm{KL}}\left[q\left(a_\tau|z_\tau\right)\|p\left(a_\tau\right)\right]\right] \\ &= \mathcal{G}_{\mathrm{CL}}(\tau). \end{aligned} \tag{8}$$

If the prior of action $p(a_t)$ represents the expert's action distribution, the action value term in the contrastive EFE $\mathcal{G}_{\mathrm{CL}}(\tau)$—the KLD between the action's predictive distribution $q(a_t|z_t)$ and the prior of action $p(a_t)$—acts as a constraint to align the inferred actions with the demonstrations. This enables contrastive EFE $\mathcal{G}_{\mathrm{CL}}(\tau)$ to be used as the LfD objective.

B. FRAMEWORK

The proposed framework consists of a world model, an action model, and an EFE model. The world model is learned first, and subsequently, with the world model's parameters fixed, the action model and EFE model are learned simultaneously. During the training of the action model and the EFE model, their losses are designed not to influence each other's parameters. The overview of the proposed framework is shown in Fig. 1.

1) World model

We utilize the recurrent state-space model (RSSM) [13], [40], [61] shown in Fig. 2 for the world model. RSSM has its own hidden states z_t, which is the concatenation of the deterministic states d_t and the stochastic states s_t at time step $t \in [1,T]$.

The deterministic states d_t are calculated from the hidden states z_{t-1} and action a_{t-1} at the previous time step $t-1$. We define the distribution of the deterministic states $p_\theta(d_t|z_{t-1},a_{t-1})$ as follows:

$$\begin{aligned} p_\theta(d_t|z_{t-1},a_{t-1}) &= \delta(d_t - d_t') \\ d_t' &= f_\theta\left(z_{t-1},a_{t-1}\right). \end{aligned} \tag{9}$$

Here, $\delta(\cdot)$ represents the Dirac delta function and d_t' is described by the deterministic function f_θ based on the hidden state dynamics of the gated recurrent unit (GRU) [62], [63].

The stochastic states s_t are represented by a set of one-hot vectors sampled from the prior or posterior in the form of categorical distributions [13]. To optimize the model's parameters using back-propagation, the straight-through gradient [64] is employed. The prior p_θ is predicted from the deterministic states d_t. In contrast, the posterior q_θ is inferred

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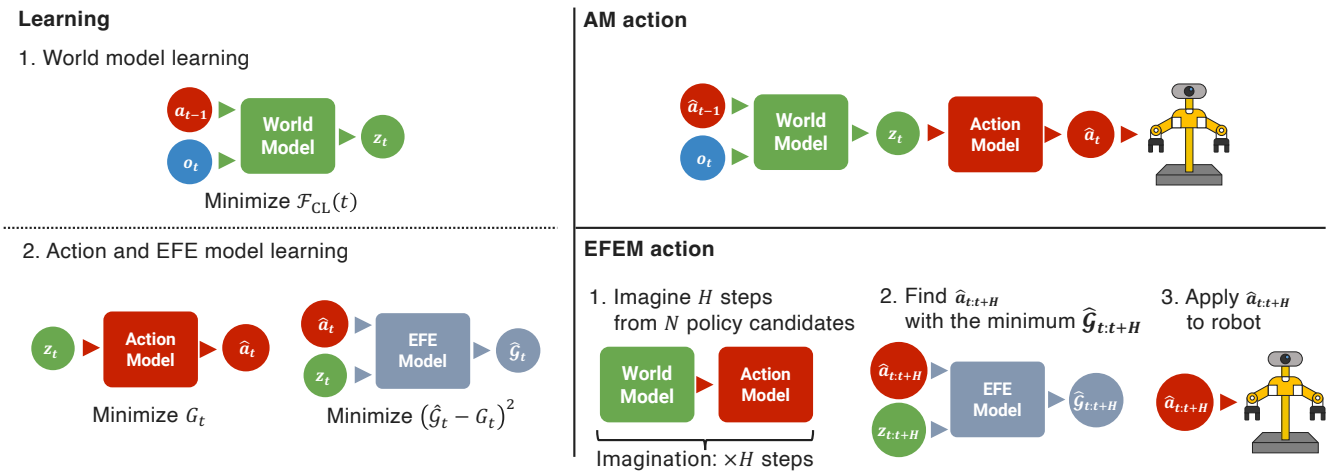

FIGURE 1. The overview of the proposed framework. For learning, the world model is learned to minimize $\mathcal{F}_{\mathrm{CL}}(t)$, as detailed in Section IV-B1. Afterwards, the action model is learned to minimize G_t, as detailed in Section IV-B2. The EFE model is simultaneously learned with the action model, as detailed in Section IV-B3. For action generation, the proposed framework has two possible way: AM action and EFEM action, as detailed in Section IV-C.

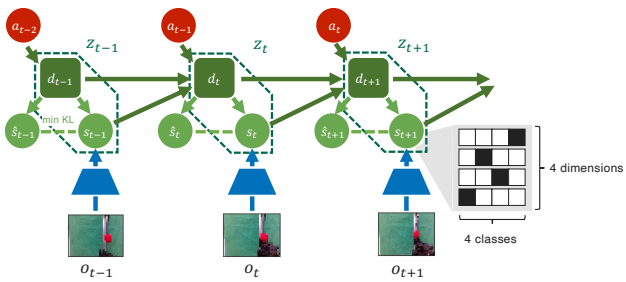

FIGURE 2. The world model. This model consists of hidden states z_t at a time step t, which comprises deterministic states d_t and stochastic states s_t. Both prior and approximate posteriors are defined for the stochastic states. The deterministic states are a 16-dimensional GRU hidden layer, and the stochastic states assume a four-dimensional categorical distribution with four classes.

from the deterministic states d_t and the observation o_t. The image encoder uses a convolutional neural network (CNN) to embed a high-dimensional visual observation o_t into the low-dimensional feature states.

In summary, the RSSM components are as follows.

$$\begin{aligned} &\text{Model states:} && z_t = \{d_t, s_t\} \\ &\text{Deterministic states:} && d_t \sim p_\theta\left(d_t | z_{t-1}, a_{t-1}\right) \\ &\text{Prior:} && \hat{s}_t \sim p_\theta\left(\hat{s}_t | d_t\right) \\ &\text{Approximate Posterior:} && s_t \sim q_\theta\left(s_t | d_t, o_t\right). \end{aligned} \tag{10}$$

The initial hidden states of the world model are obtained using a neural network named "initializer," which was designed for world model initialization. The initial observation, denoted as o_0, is input into the encoder. From the features output by the encoder, deterministic initial states d_0 are generated by the initializer. Subsequently, the world model's prior is employed to obtain initial stochastic states s_0 from the initial deterministic states d_0. The initializer is learned end-to-end with the world model.

The training of the world model is conducted by minimizing the contrastive variational free energy $\sum_t \mathcal{F}_{\mathrm{CL}}(t)$. During training, the KLD term is multiplied by a hyperparameter β. Furthermore, KL-balancing [13] is used when calculating the KLD. KL-balancing involves a hyperparameter w, which is a value between 0 and 1. The hyperparameter determines how the gradient from the KL term is distributed between the prior (given a weight of w) and the approximate posterior (given a weight of $1-w$) as follows:

$$\begin{aligned} \mathcal{F}_{\mathrm{CL}}(t) \cong \beta \cdot \big(& w \cdot D_{\mathrm{KL}}\left[\mathrm{sg}\left(q_\theta\left(s_t | d_t, o_t\right)\right) \| p_\theta\left(s_t | d_t\right)\right] \\ & + (1-w) \cdot D_{\mathrm{KL}}\left[q_\theta\left(s_t | d_t, o_t\right) \| \mathrm{sg}\left(p_\theta\left(s_t | d_t\right)\right)\right]\big) \\ & - I_{\mathrm{NCE}}\left(Z_t; O_t\right). \end{aligned} \tag{11}$$

Here, $\mathrm{sg}(\cdot)$ denotes a function that makes input independent of back-propagation. To compute $I_{\mathrm{NCE}}\left(Z_t; O_t\right)$, we employ a multilayer perceptron (MLP) for the critic $f_\theta(o, s) = h_\theta(o)^T g_\theta(z)$ analogously to previous works [25], [48].

2) Action model

The action model q_ϕ is implemented as an MLP with two hidden layers. The action a_t at a time step t is inferred as follows:

$$a_t \sim q_\phi(a_t | z_t). \tag{12}$$

We assume a Gaussian distribution for the action's predictive distribution $q_\phi(a_t | z_t)$. Therefore, the inputs to the action model are the hidden states z_t, and the output consists of the mean and standard deviation. To optimize the model's parameters using back-propagation, the reparameterization trick [54], [55] is employed. However, the action range is constrained by applying the hyperbolic tangent function to the mean.

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The actual learning objective of the action model $\sum_t G_t$ involves the use of generalized advantage estimation [65], as in the original framework [25]:

$$G_t = \mathcal{G}_{\mathrm{CL}}(t) + \gamma \begin{cases} (1-\lambda) g_\psi(z_{t+1}, a_{t+1}) + \lambda G_{t+1} & \text{if } t < T \\ \hat{\mathcal{G}}_T & \text{if } t = T \end{cases}. \tag{13}$$

Here, λ, γ are hyperparameters, T is the horizon for trajectories, and $g_\psi(\cdot)$ is the EFE model. This allows for an approximation of the EFE $\mathcal{G}_{\mathrm{CL}}(t)$ at each time step, thereby enabling computing further into the future in a recursive manner.

3) EFE model

The EFE model g_ψ is implemented as an MLP with one hidden layer. We model to predict the EFE $\hat{\mathcal{G}}_t$ at a time step t as follows:

$$\hat{\mathcal{G}}_t = g_\psi\left(z_t, a_t\right). \tag{14}$$

We predict EFE from both the hidden states z_t and the action a_t, as in previous studies [20], [43], [44]. Using the EFE model, we can estimate the expectation of EFE, thereby replacing the calculation of the actual EFE. In this case, the EFE model is trained by taking the G_t from (13) and minimizing the squared error, as follows:

$$\mathcal{L}_\psi = \sum_t \left(\hat{\mathcal{G}}_t - G_t\right)^2. \tag{15}$$

C. ACTION GENERATION

The proposed framework allows for two types of action generation. The first is the "action model–based action (AM action)." In this approach, we utilize the action model $q_\phi(a_t|z_t)$ to generate action a_t, which depends only on the current hidden states z_t. AM action is a commonly used method in active inference with deep learning [23], [25]. However, it may not fully leverage the benefits of active inference [12], [17], which selects a policy that minimizes EFE for adaptive action.

Therefore, we propose a second approach: "EFE model–based action (EFEM action)." In this approach, we select a policy with length H, which minimizes the estimated value of the EFE model from N policy candidates. This approach is more in line with active inference that selects actions adaptively. In summary, the detailed EFEM action algorithm is shown in Algorithm 1. The number of policies to generate N as well as the prediction steps H can be adjusted as hyperparameters depending on the robot's computational capabilities.

Algorithm 1 EFEM action Algorithm

Initialize hyperparameters
N = Number of policy samples
H = Prediction horizon
for $i = 1$ to N **in parallel do**
  for $k = 0$ to H **do**
    Generate action using the action model:
      $a^i_{t+k} \sim q_\phi(a_t|z^i_{t+k})$
    Predict next states using the world model:
      $d^i_{t+k+1} = f_\theta(z^i_{t+k}, a^i_{t+k})$
      $\hat{s}^i_{t+k+1} \sim p_\theta(\hat{s}^i_{t+k+1}|d^i_{t+k+1})$
      $z^i_{t+k+1} = \{d^i_{t+k+1}, \hat{s}^i_{t+k+1}\}$
  end for
  Calculate EFE estimate using EFE model: $\hat{\mathcal{G}}^i_{t:t+H} = g_\psi(z^i_{t:t+H}, a^i_{t:t+H})$
end for
Find policy $a^{\mathrm{argmin}(\hat{\boldsymbol{\mathcal{G}}}^i_{t:t+H})}_{t:t+H}$

V. EXPERIMENT

A. ENVIRONMENT SETUP

1) Robot

For the experiments, the Rakuda-2 robot (Robotis Japan) was used (Fig. 3). This robot has a total of 17 degrees of freedom and five fixed joints (two related to the viewpoint, one for the body's rotation, and both arm grippers). Thus, the data were obtained with the robot having 12 degrees of freedom. The robot's head is equipped with a camera (RealSense Depth Camera D435; Intel) capable of capturing RGB images at a resolution of 360×640.

2) Task setting

To demonstrate the scalability of the proposed framework in a real-world environment, we created a ball-reaching task. In our experimental setup (Fig. 3), a red ball is suspended from a pillar. The initial environment involves both of the robot's arms in a lowered position. The task's preferred observation, denoted as $\tilde{o}$, is defined as the observation where the gripper touches the suspended ball. The ball was placed in 16 different positions: four in the horizontal direction and four in the vertical direction (Fig. 4). The robot must use the appropriate arm to touch the ball based on its position. For each of these positions, reaching demonstrations were recorded 20 times, resulting in a total of 320 demonstrations, by teleoperating the robot in a leader–follower manner. Therefore, each demonstration involved slightly different actions, even though the ball position was the same. The collected data include the robot's joint angles and visual images, recorded at 50 time steps at a frequency of 5 Hz. The resolution of the RGB images was originally 360×640 but this was downsized to 24×32 for use in the experiment as observation o_t. Here, the prior of action $p(a_\tau)$ is modeled as a Gaussian distribution with a time-variant mean and 0.1 standard deviation, characterized by the demonstration data $\tilde{a}_t$ included in the training set:

$$p\left(a_t\right) \approx \mathcal{N}\left(\tilde{a}_t, 0.01\right). \tag{16}$$

In this experiment, actions a_t were replaced with joint angles j_t. The pixel values in the images were normalized within the range of 0 to 1. Additionally, the joint angles were normalized

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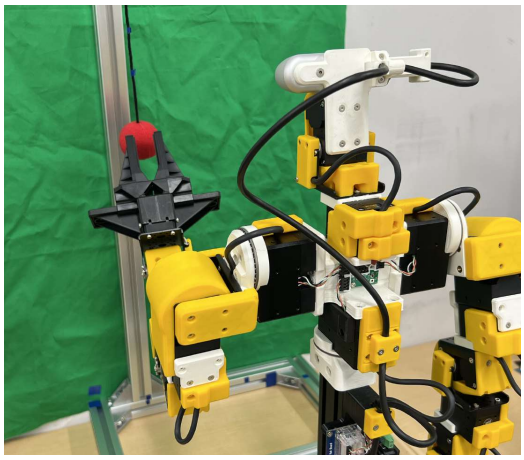

FIGURE 3. The experimental environment. This environment consists of a red ball, a pillar, and a dual-arm robot (Rakuda-2; Robotis Japan). The ball is suspended from the pillar. Initially, the robot's arms are in a lowered position. The position of the ball can be adjusted to any location by changing the pillar's position and the length of the suspending string. Additionally, a green fabric serves as the background.

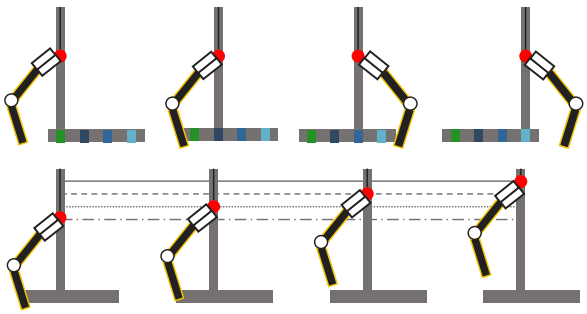

FIGURE 4. Positions of the ball in the demonstrations. The data consist of 16 different positions where the red ball was placed, with 4 positions in the horizontal direction and 4 positions in the vertical direction. In cases where the ball was placed on the left side in the horizontal direction, the robot's left arm was used to make contact with the ball. Similarly, when the ball was placed on the right side in the horizontal direction, the robot's right arm was used to make contact with the ball.

within the range of -0.95 and 0.95 to avoid gradient vanishing in the hyperbolic tangent function of the action model.

B. BASELINES

We compared our proposed framework with three recent promising (non-free-energy-based) LfD baselines: conditional neural movement primitives (CNMP) [14], conditional neural expert processes (CNEP) [15], and implicit behavior cloning (IBC) [16]. CNMP receives a current time representation and observation as well as a target time representation to generate corresponding actions from a query network. CNEP is an extension of CNMP, which equips multiple query networks and a gating network that selects the best query network. IBC is based on energy-based models [66], which generates actions by inferring actions with the minimum energy from current observations. We implemented these frameworks as variants with CNNs to encode the ball positions based on observations. In particular, for IBC, we provided the robot with an exponential moving average of the IBC's predicted actions to guarantee continuous action generation.

C. ABLATIONS

We compare our proposed framework with two other ablation frameworks. We refer to our proposed framework as "contrastive+categorical," one of the ablation frameworks as "likelihood+categorical," and the other as "contrastive+Gauss."

Likelihood+categorical

Setting hyperparameter α in (4) and (7) to 1 corresponds to adopting the original likelihood-based definition of free energy. In this case, a likelihood-based method uses image reconstruction, and the world model includes a decoder $p_\theta(o_t|z_t)$, which corresponds to the likelihood function for observations. This decoder utilizes a Transposed CNN, taking the world model states z_t as input and reconstructing it into an observation o_t.

Contrastive+Gauss

Although the proposed method assumed a categorical distribution as the stochastic states of the world model, it is possible to assume a Gaussian distribution instead for the stochastic states [25], [39], [40]. In this case, the calculated mean of the Gaussian distribution is bounded by the hyperbolic tangent function. To optimize the model's parameters using back-propagation, the reparameterization trick [54], [55] is employed.

D. EXPERIMENTAL CRITERIA

We evaluated the proposed framework in real-world robot settings, using high-dimensional image-based data. Our experiment aims to answer the following questions: (i) Is it possible to achieve high-dimensional goals in a real-world environment with the proposed framework? (ii) Does the framework perform better than the baselines? (iii) Does the proposed framework perform better than the ablation frameworks? (iv) Is the EFE model accurate enough to estimate the actual calculation? (v) Can the EFEM action improve the performance in an unlearned environment?

1) Static ball position

To examine questions (i), (ii), and (iii), we conducted three different cases: a learned position, an unlearned position, and a background change. We evaluated the performance of the proposed and ablation frameworks in all three cases, but only the baselines in the learned-position case. In all three experiments, both the proposed and ablation frameworks generated actions using AM action and used the expectation of the action's predictive distribution $q_\phi(a_t|z_t)$ (i.e., the mean value of the distribution) as the actual action.

In the learned-position case, the experiment involved placing the balls in the 16 learned positions. Ten trials were conducted at each of the 16 positions, for a total of 160 trials.

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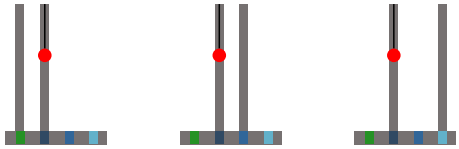

FIGURE 5. Example of the background-change case. When we placed the ball in the second column from the left, there were three different ways to add an additional pillar.

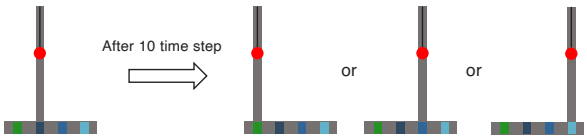

FIGURE 6. Environmental change in EFEM action experiment. In this experiment, the balls were initially placed in the 16 learned positions. However, after the robot began generating actions, at around 10 time steps, the balls were moved to different trained positions. In this case, the vertical positions remained the same, but the ball was positioned at one of three different positions in the horizontal direction.

Algorithm 2 EFEM action in this experiment

Initialize the hidden states of the world model:
  z_0 = initializer(o_0)
Perform EFEM action from z_0
and get the imagined best policy $\hat{a}_{0:5}$
for $t = 1$ to T **do**
  Recieve observation: o_t
  Predict next states using the world model:
    Get deterministic states: $d_t = f_\theta(z_{t-1}\hat{a}_{t-1})$
    Sample stochastic states: $s_t \sim q_\theta(s_t|d_t, o_t)$
    Concatenate states: $z_t = \{d_t, s_t\}$
  if $t \bmod P = 0$ **then**
    Perform EFEM action from z_t
    and get the imagined best policy $\hat{a}_{t:t+H}$
  end if
  Apply the imagined action $\hat{a}_t$ to robot
end for

In the unlearned-position case, the balls were placed in the positions between the 16 learned positions both vertically and horizontally, which were not included in the demonstration. Five trials were conducted at each of these unlearned positions, for a total of 120 trials.

In the background-change case, we introduced an additional pillar to the environment, which occupied a significant portion of the visual information but was not essential to the task and therefore may affect the model's decision. We evaluated how such a significant object affects the ball position perception between contrastive and likelihood-based learning by comparing the success rates of the proposed framework and ablations. Three possible locations for adding a pillar were selected (Fig. 5) and five trials were conducted at each of these positions, for a total of 240 trials.

We also evaluated the generalization capabilities of our proposed framework in several other cases, as detailed in the Appendix. C.

Moreover, to examine question (iv), we assess the estimated values from the proposed framework's EFE model for the scenarios of success and failure in the learned position case.

2) Dynamic ball position

To examine question (v), experiments were conducted in a dynamic environment, using the proposed framework and both AM and EFEM actions. In this experiment, the balls were initially placed in one of the 16 learned positions. However, after the robot began generating actions, at around 10 time steps, the balls were moved to different learned positions, as shown in Fig. 6. For each of these new positions, five trials were conducted, which means that there were 15 trials for each of the 16 initial ball positions, for a total of 240 trials.

Ideally, the robot should perform EFEM actions at every time step and use the first action $\hat{a}_t$ from the obtained policy $\hat{a}_{t:t+H}$. However, this is difficult to achieve due to computational costs. Therefore, the EFEM action was performed at the beginning of the task and with a period of P steps. Accordingly, it is necessary to replace the actions at time steps where EFEM actions are not performed with the actions from the latest EFEM action policy $\hat{a}_{t+1:t+H}$. The detailed EFEM action process is shown in Algorithm 2.

The prediction horizon H was set to 50 at the beginning of the task and 5 with a period of $P = 5$ steps. The number of policies generated, denoted as N, was initially 500 and was changed to 40 with a period of $P = 5$ steps. In this case, as the environment underwent dynamic changes, we aimed to promote the generation of policies corresponding to the altered states. To achieve this, we performed EFEM actions with N policies, with half based on the states from the previous world model, and the other half derived from states generated by an initializer. To facilitate the EFEM action, it is necessary to generate multiple distinct actions, which are sampled from the action's predictive distribution $q_\phi(a_t|z_t)$. Additionally, to ensure consistency, actions sampled from action's predictive distribution $q_\phi(a_t|z_t)$ are also used for AM action. Moreover, to introduce variability in the actions, we intentionally adjusted the standard deviation of the action's predictive distribution. At the beginning of the task, the standard deviation was 0.001, and it was changed to 2.5 with a period of $P = 5$ steps. This technique was also applied to both AM and EFEM actions.

VI. RESULTS AND DISCUSSIONS

A. STATIC BALL POSITION

1) Success rates

The success rates of the proposed method were compared with the baselines and the ablation frameworks in each case (Fig. 7). We conducted statistical analysis of the data for each case with a significance level of 0.05. First, Cochran's Q test [67] was used to examine differences among the methods, and if significant differences were found, Ryan's multiple comparison test [68] for proportions was subsequently applied to perform detailed comparisons between the methods.

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In the learned-position case, the test revealed a significant difference among the methods ($Q = 254.75, p < 0.001$), and subsequent multiple comparisons showed that the success rate of the proposed "contrastive+categorical" was significantly higher than those of the other methods. CNMP and CNEP struggled to generate precise action and failed to reach the balls. The low success rate of IBC was arguably due to the robot's initial posture and observation. Initially, the robot was not able to see its arms in its vision because both of the robot's arms were in a lower position. Therefore, the failure of perceiving the initial robot's arm positions resulted in IBC keeping the robot's arms out of the robot's vision. Conversely, our proposed framework reached the balls precisely by perceiving environment situation through the world model, as EFE involves both LfD term and goal-oriented contrastive learning term. These results suggest the importance of both perception using the world model and action to minimize EFE even in a learned environment. Moreover, the proposed "contrastive+categorical" also outperformed both the ablation frameworks. This indicates that the use of contrastive learning and categorical distributions improved the success rate.

In the unlearned-position case, the test revealed a significant difference among the methods ($Q = 41.49, p < 0.001$), and subsequent multiple comparisons showed that the success rate of the proposed framework was significantly higher than those of the other ablation frameworks. The decrease in success rates compared to those in the learned-position case was due mainly to challenges in the approach using the world and action models. The world model must generalize to unlearned ball positions, and the action model must generate actions for unfamiliar world model states. The two-fold generalization process in the experimental setup makes success difficult, resulting in significant rate decreases. Further discussion is provided in Section VI-A2.

In background-change case, the test revealed a significant difference among the methods ($Q = 99.61, p < 0.001$), and subsequent multiple comparisons showed that the success rate of the proposed framework was significantly higher than those of the ablation frameworks. The reduced success rate with the additional pillar may be due to the fact that the pillar provides easily interpretable information and interferes with proper perception. For instance, the ball's horizontal position might be clearer when observing the pillar than when observing the ball directly. Notably, the success rates of "contrastive+categorical" and "contrastive+Gauss" outperformed "likelihood+categorical" because of the contrastive learning, making the models less concerned about the additional pillar of the background. The contrastive learning makes the models less sensitive to background features, as demonstrated in prior research [24], [25], [51].

2) Analysis of latent state representation

To more thoroughly investigate why the proposed framework performed the best, we performed an analysis of the hidden states in the world model of both the proposed and the ablation frameworks. We compared the deterministic states d_t of the world model in each framework because the stochastic states representations differ in each framework. Note that for this analysis, we used the states inferred from the approximate posterior $q_\theta(s_t|d_t, o_t)$ by providing visual observations o_t and actions a_t from demonstrations, where each demonstration corresponds to a different 16 ball positions.

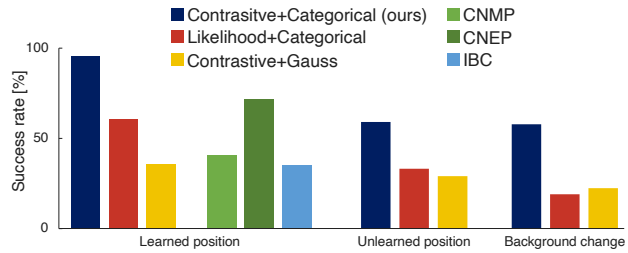

FIGURE 7. Comparison of success rates across the proposed method, the baselines, and the ablations in each case. The name of each column indicates the experimental setting in which the success rates were obtained.

As shown in Fig. 8, the proposed "contrastive+categorical" framework provides a clear and easily distinguishable representation of states for differences in ball positions as well as changes over time during the task. The framework yields meaningful state representations for achieving the task. In contrast, the "likelihood+categorical" and "contrastive+Gauss" frameworks tend to produce similar states for different ball positions and the task progresses. Therefore, we can conclude that the proposed framework encourages the acquisition of better state representations, resulting in a higher success rate in performing the task compared with the ablation frameworks. An example of latent states in the unlearned-position case is shown in Fig. 8. In this example, the ball is positioned between positions 1 and 5. Given that the latent states are well-defined according to the ball's position, this analysis indicates that the world model can generalize to infer unlearned ball positions, suggesting that the bottleneck in generalization primarily resides within the action model.

3) EFE estimation

We demonstrate the capability of the EFE model to estimate appropriate EFE values for the scenarios of success or failure. Examples at these instances are illustrated in Fig. 9. In the case of a successful task, the estimated EFE values $\hat{\mathcal{G}}_t$ indicate that the further the states are from a preferred state, the higher the EFE value. Conversely, as the states approach a preferred state, the EFE value decreases. In the case of a failure, the EFE is estimated to decrease until around time step 15. However, after this point, the estimated value starts to increase. This indicates that the model predicts a high EFE when the robot reaches a place far from the preference. From these results, we can conclude that the EFE model is capable of estimating low free energy for states and actions that lead to achieving the preference as well as high free energy for states and actions that are unlikely to achieve the preference.

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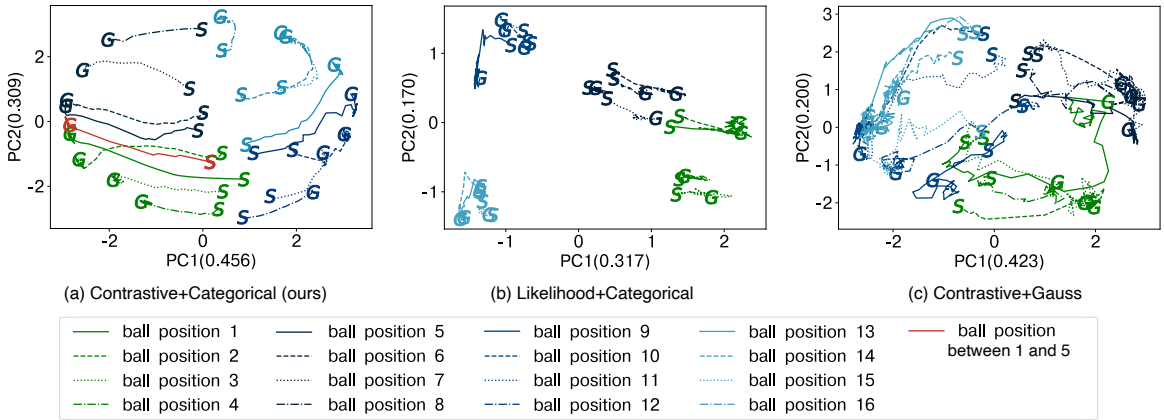

FIGURE 8. **Deterministic states of the world model. We performed principal component analysis and plotted the first and the second principal components of the deterministic states. The values in the parentheses next to each axis indicate the explained variance ratio. The points denoted as "S" and "G" represent the states of the initial and final time steps, respectively. The color and line style of the ball's position correspond to the different positions: the same color indicates the same vertical position, and the same line style indicates the same horizontal position in Fig. 4, except the red solid line, which is an example of latent states in the unlearned-position case, in which the ball was positioned between positions 1 and 5.**

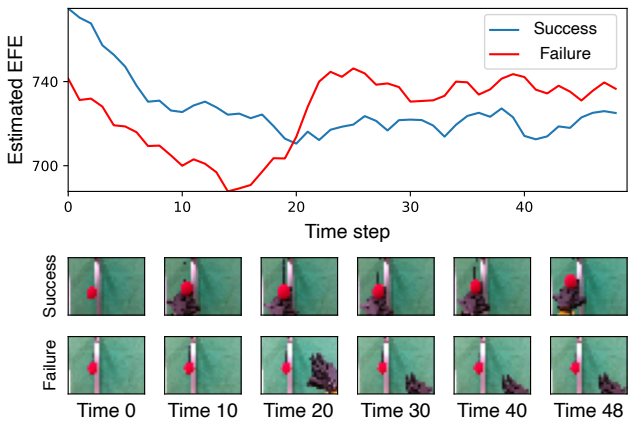

FIGURE 9. **Estimated EFE for the scenarios of success and failure. The images represent visual information at corresponding time steps. The estimates shown are obtained by applying a weighted moving average with a weight of 0.8.**

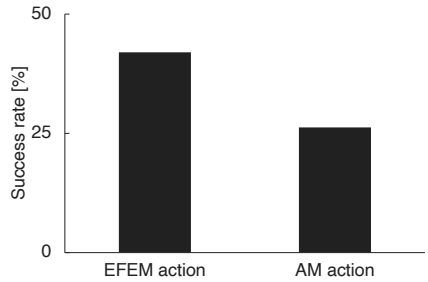

FIGURE 10. **Comparison of success rates between EFEM and AM actions. The name of each column indicates the action method. EFEM action significantly outperformed AM action.**

B. DYNAMIC BALL POSITION

McNemar's test [69] revealed that the success rate of the EFEM action was significantly higher than that of the AM action (McNemar's $\chi^2 = 63.0$, $p = 0.001$) (Fig. 10). The substantial difference in success rates can be attributed to the movement of the ball from the left to the right side and vice versa. When the ball moves between the two left positions or the two right positions, the robot can reach the ball by simply adjusting the arm it has already raised. However, when the ball moves from the left side to the right or vice versa, the robot needs to raise the arm on the opposite side for reaching. Only EFEM action methods enabled this recovery from failure by switching arms because it considered the estimated EFE when choosing actions, as shown in Fig. 11. This result suggests that considering EFE online during action generation outperforms training models with EFE as the learning objective.

VII. THREATS TO VALIDITY

Internal validity

In this experiment, the ball position was set by the experimenter adjusting the strings and pillars from which the ball was suspended. Therefore, during the experiment, even though the setting was supposed to be the same, it was possible that the ball is not positioned exactly the same each time. To mitigate the impact of this variation, multiple trials were conducted for each setting.

External validity

We showed the performance of our proposed framework in static and dynamic environments, including learned and unlearned ball positions and background changes. However, the other settings, such as background clutter, lighting conditions, object shape, or extrapolated positions, may affect the performance of the proposed framework. To investigate the influence of these settings on the performance, we conducted additional experiments, as detailed in Appendix C.

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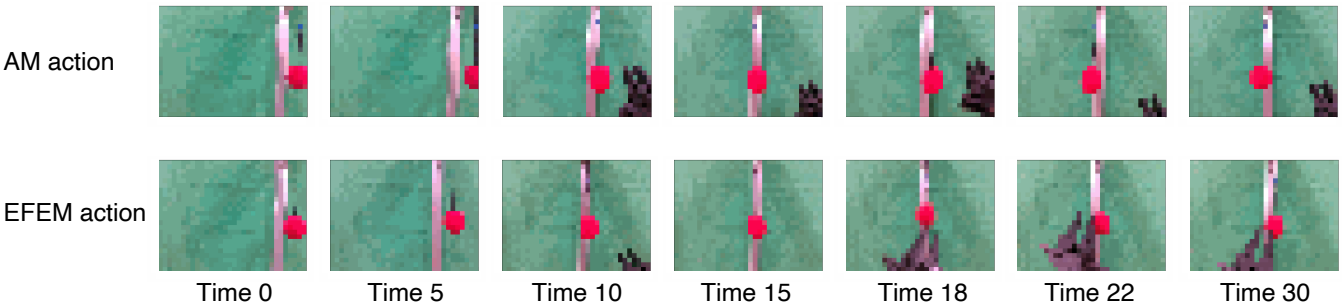

FIGURE 11. Example of behavior when the ball moves. When the ball's position changed, the EFEM action was able to reach for the ball by lifting the arm opposite to the one previously raised. However, such behavior was not observed with AM action.

Construct and conclusion validity

We chose Cochran's Q test [67], Ryan's multiple comparison test [68], and McNemar's test [69] for proportions as the statistical analysis methods for the results. Since these analyses are used to test for differences between groups of proportions when there is correspondence, they are reasonable methods for comparing success rates between methods with correspondence at the ball position. However, in this study, the reaching task was judged as a success when the arm made contact with the ball. Given that this binary judgment classifies both near-misses and obvious failures as failures, it is possible that other judgment methods could reveal new important trends. Furthermore, we only conducted simple robotic reaching tasks. Therefore, whether the significant superiority of the proposed framework is maintained in other complex tasks remains to be verified.

VIII. CONCLUSION

In this study, we proposed a framework that extends contrastive active inference by incorporating LfD for real-world robot control. We introduced a hyperparameter α that controls the upper bound on surprise and demonstrated that it yields both contrastive and original free energy. We also proposed an action-selection method based on estimated EFE. We conducted experiments comparing our proposed framework with the baselines and ablations in both learned and unlearned environments, confirming its applicability. The experimental results showed that our framework enabled task accomplishment in both learned and unlearned environments. The results also highlighted the potential for improving performance by using contrastive free energy–based learning and by modeling states based on categorical distribution within the world model. Furthermore, we demonstrated that selecting actions based on estimated EFE leads to more adaptive behaviors in dynamic environments.

The proposed framework did not achieve a good success rate in unlearned environments compared with learned environments, particularly when the ball's position was unlearned. To address these challenges, it might be beneficial to explore learning from human curiosity-driven behaviors [6], [33], [34]. This might potentially enable the model to adapt to a wider range of situations.

APPENDIX A CONTRASTIVE ACTIVE INFERENCE

In its original derivation, the contrastive variational free energy $\mathcal{F}_{\mathrm{CL}}(t)$, is obtained by adding log-probability $\log p(o)$ to the original variational free energy $\mathcal{F}(t)$ [25] as follows:

$$\begin{aligned}\mathcal{F}_{\mathrm{CL}}(t) &\stackrel{+}{=} D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t|o_t\right)\right] \\ &\quad - \log p\left(o_t\right) + \log p\left(o_t\right) \\ &= D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t\right)\right] \\ &\quad - E_{q(z_t|o_t)}\left[\log p\left(z_t|o_t\right) - \log p\left(z_t\right)\right] \\ &= D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t\right)\right] - I_{\mathrm{NCE}}\left(Z_t; O_t\right).\end{aligned} \tag{17}$$

The contrastive EFE $\mathcal{G}_{\mathrm{CL}}(\tau)$ for the future is obtained by adding the log-prior preference distribution $\log \tilde{p}(o_\tau)$ to the original EFE $\mathcal{G}(\tau)$ [25]:

$$\begin{aligned}&\mathcal{G}_{\mathrm{CL}}(\tau) \\ &\stackrel{+}{=} E_{\tilde{p}(o_\tau)q(z_\tau,a_\tau)}\left[\log q\left(z_\tau, a_\tau\right) - \log \tilde{p}\left(o_\tau, z_\tau, a_\tau\right) + \log \tilde{p}\left(o_\tau\right)\right] \\ &= D_{\mathrm{KL}}\left[q\left(z_\tau\right)\|p\left(z_\tau\right)\right] \\ &\quad - I_{\mathrm{NCE}}\left(Z_\tau; \tilde{O}_\tau\right) \\ &\quad + E_{q(z_\tau,a_\tau)}\left[q\left(a_\tau|z_\tau\right) - p\left(a_\tau\right)\right].\end{aligned} \tag{18}$$

In this framework, the authors further assume that when considering an action, the agent has a good approximation of the prior in its generative model, that is, $D_{\mathrm{KL}}\left[q\left(z_\tau\right)\|p\left(z_\tau\right)\right] = 0$ [25], [60]. Then, the contrastive EFE $\mathcal{G}_{\mathrm{CL}}(\tau)$ becomes as follows:

$$\mathcal{G}_{\mathrm{CL}}(\tau) \approx -I_{\mathrm{NCE}}\left(Z_\tau; \tilde{O}_\tau\right) - E_{q(z_\tau,a_\tau)}\left[q\left(a_\tau|z_\tau\right) - p\left(a_\tau\right)\right]. \tag{19}$$

In other words, the agent aims to maximize the mutual information between the future hidden states z_τ and the preferred observation $\tilde{o}$, that is, to choose an action a_τ that leads to hidden states z_τ that is likely to be the preferred observation $\tilde{o}$ by minimizing the contrastive EFE $\mathcal{G}_{\mathrm{CL}}(\tau)$.

The contrastive free energy was derived by adding $\log p(o_t)$ or $\log \tilde{p}(o_t)$ [25]. We suggest modifying the derivation of contrastive free energy to be more closely coupled to the original free energy and the surprise. Accordingly, we derive the contrastive free energy from the surprise upper bound.

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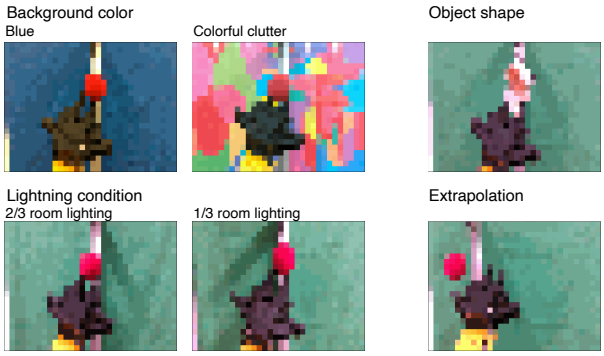

FIGURE 12. Additional experimental environments. Each figure shows an example of the observation in the corresponding environments.

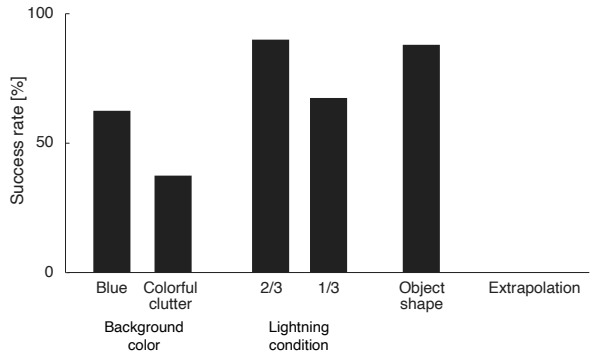

FIGURE 13. Comparison of success rates across the additional experimental environments. The name of each column indicates the specific experimental environment. Note that the success rate in the "Extrapolation" environment is 0%.

APPENDIX B SURPRISE UPPER BOUND

Here is the detailed derivation of the contrastive variational free energy $\mathcal{F}_{\mathrm{CL}}(t)$ from the surprise upper bound $F_\alpha(t)$:

$$\begin{aligned} F_{\alpha\gg1}(t) &\approx D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t|o_t\right)\right] \\ &= E_{q(z_t|o_t)}\left[\log q\left(z_t|o_t\right) - \log p\left(z_t|o_t\right)\right] \\ &\quad - \log p\left(z_t\right) + \log p\left(z_t\right) \\ &= E_{q(z_t|o_t)}\left[\log q\left(z_t|o_t\right) - \log p\left(z_t\right)\right] \\ &\quad - E_{q(z_t|o_t)}\left[\log p\left(z_t|o_t\right) - \log p\left(z_t\right)\right] \\ &\le D_{\mathrm{KL}}\left[q\left(z_t|o_t\right)\|p\left(z_t\right)\right] - I_{\mathrm{NCE}}\left(Z_t;O_t\right) \\ &= \mathcal{F}_{\mathrm{CL}}(t). \end{aligned} \tag{20}$$

Here, the upper bound in the fourth line is from the NCE mini-batch bound [24], [49].

We also show the detailed derivation of the contrastive expected free energy $\mathcal{G}_{\mathrm{CL}}(\tau)$ from $G_\alpha(\tau)$. In this derivation, we assume $D_{\mathrm{KL}}\left[q\left(z_\tau\right)\|p\left(z_\tau\right)\right] = 0$, as in the original derivation [25]. Moreover, $E_{\tilde{p}(o_\tau)q(z_\tau)}\left[\log p\left(z_\tau|o_\tau\right) - \log p\left(z_\tau\right)\right]$ is lower bounded by $I_{\mathrm{NCE}}\left(Z_\tau;\tilde{O}_\tau\right)$ [24], [25], [49]. Accordingly, the derivation is as follows:

$$\begin{aligned} G_{\alpha\gg1}(\tau) &\cong \mathcal{G}_{\mathrm{CL}}(\tau) \\ &= E_{\tilde{p}(o_t)}\left[D_{\mathrm{KL}}\left[q\left(z_t,a_t\right)\|p\left(z_t,a_t|o_t\right)\right]\right] \\ &= E_{\tilde{p}(o_t)q(a_t,z_t)}\left[\log q\left(a_t|z_t\right) + \log q\left(z_t\right)\right. \\ &\quad \left. - \log p\left(z_t|o_t\right) - \log p\left(a_t\right)\right] \\ &\quad + E_{\tilde{p}(o_t)q(a_t,z_t)}\left[\log p(z_t) - \log p(z_t)\right] \\ &= E_{q(z_t)}\left[\log q\left(z_t\right) - \log p\left(z_t\right)\right] \\ &\quad - E_{\tilde{p}(o_t)q(z_t)}\left[\log p\left(z_t|o_t\right) - \log p\left(z_t\right)\right] \\ &\quad + E_{q(a_t,z_t)}\left[\log q\left(a_t|z_t\right) - \log p\left(a_t\right)\right] \\ &\approx -E_{\tilde{p}(o_\tau)q(z_\tau)}\left[\log p\left(z_\tau|o_\tau\right) - \log p\left(z_\tau\right)\right] \\ &\quad + E_{q(a_\tau,z_\tau)}\left[\log q\left(a_\tau|z_\tau\right) - \log p\left(a_\tau\right)\right] \\ &\le -I_{\mathrm{NCE}}\left(Z_\tau;\tilde{O}_\tau\right) + E_{q(z_\tau)}\left[D_{\mathrm{KL}}\left[q\left(a_\tau|z_\tau\right)\|p\left(a_\tau\right)\right]\right] \\ &. \end{aligned} \tag{21}$$

APPENDIX C ADDITIONAL EXPERIMENTS

We show the limitations of our proposed framework in the unlearned-position and background-change cases, as discussed in Section VI-A. However, other changes in environmental conditions may affect the proposed framework. To demonstrate the generalization capabilities of the proposed framework, we conducted additional experiments with diverse environmental change cases: two background colors, two lighting conditions, a object shape, and an extrapolation. In all cases, the proposed framework generated actions using AM action and used the expectation of the action's predictive distribution $q_\phi(a_t|z_t)$ (i.e., the mean value of the distribution) as the actual action. Moreover, excluding the extrapolation case, the experiment involved placing the balls in the 16 learned positions. Five trials were conducted at each of the 16 positions, for a total of 80 trials.

Background color

We considered the background change only with an additional distracting pillar in Section VI-A. However, an unlearned background color may also affect the ball position perception. Therefore, we conducted experiments in an environment with a blue and a colorful cluttered background, as shown in Fig. 12. As shown in Fig. 13, the success rate was less than approximately two-thirds when there was a change in background color compared to the learned environment. This indicates that changes in pixel values over a large portion of the visual field can deteriorate the perception of the proposed framework. However, even in the colorful clutter environment, where it initially seems difficult to identify the ball's position, the proposed framework achieved a success rate of over one-third, indicating a certain degree of robustness.

Lightning condition

The lighting condition may also affect the performance of the proposed framework. Therefore, we conducted experiments under two lighting conditions: room lighting set to $2/3$ and $1/3$ of its normal brightness, as shown in Fig.12. As shown in Fig. 13, the proposed framework still achieved a high success rate in the environment with the room lighting set to $2/3$. When the room lighting was set to $1/3$, the proposed framework began to fail to reach the ball. These findings

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indicate that the proposed framework has better robustness to lighting conditions compared to changes in ball position and background.

Object shape

To investigate whether the proposed framework can achieve the reaching task even with an unlearned object, we conducted an experiment in an environment where a small white doll was suspended as a reaching target, as shown in Fig. 13. The proposed framework achieved a high success rate even with a target of such complex shape, as shown in Fig. 13. This result suggests that the proposed framework can be well generalized to the color and shape of the target object.

Extrapolation

We investigated how our proposed framework performs in the extrapolation case. Balls placed at the top, bottom, and right sides of the learned positions were mostly at the edge of the robot's field of view, making it difficult to see the ball when extrapolated. Therefore, we focused on left extrapolation. The ball was placed in four vertical locations, similar to the learned-position case. Five trials were conducted at each of the four positions, for a total of 20 trials.

The proposed framework was unable to reach the extrapolated positions. However, while it was challenging to reach the correct horizontal positions, there were instances where the vertical positions were correct. Combining the proposed framework with behavior generation methods that can adapt to extrapolated positions might resolve this limitation.

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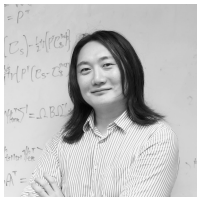
KENTARO FUJII received a B.E. from the Department of Electronics and Electrical Engineering at Keio University in 2024. He is presently pursuing a master's degree at Keio University. His research interests include cognitive robotics and robot learning.

Mr. Fujii is a student member of the Japanese Society for Artificial Intelligence and the Information Processing Society of Japan.



TAKUYA ISOMURA received his Ph.D. (Science) from the University of Tokyo in 2017.

He worked as a postdoctoral researcher at the Lab for Neural Computation and Adaptation at the RIKEN Center for Brain Science (Saitama, Japan). Since 2020, he has been a Unit Leader of the Brain Intelligence Theory Unit at the same institute. He has conducted computational research aimed at revealing the neuronal mechanisms underlying intelligence and develop biomimetic schemes for artificial intelligence using mathematical analyses and experimental validations.

Dr. Isomura is a member of the Society for Neuroscience, the Japan Neuroscience Society, and the Japanese Neural Network Society.

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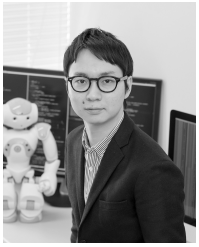

SHINGO MURATA (M'16) received his B.E., M.E., and D.E. from Waseda University in 2011, 2013, and 2016, respectively.

From 2016 to 2018, he served as a Research Associate in the Department of Modern Mechanical Engineering at Waseda University. Then, from 2018 to 2020, he held the position of Assistant Professor at the Principles of Informatics Research Division, National Institute of Informatics (NII). During this period, he was appointed as an Assistant Professor in the Department of Informatics, The Graduate University for Advanced Studies (SOKENDAI). From 2020 to 2024, he was a Senior Assistant Professor in the Department of Electronics and Electrical Engineering at Keio University. Presently, he is an Associate Professor in the same department. His research interests presently include cognitive robotics, robot learning, and computational psychiatry.

Dr. Murata is a member of the Robotics Society of Japan and the Japanese Society for Artificial Intelligence. He has served as an Associate Editor of *IEEE Transactions on Cognitive and Developmental Systems* since 2021. He received the Best Paper Award at the 25th International Conference on Artificial Neural Networks in 2016.

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